

ALLAN PENG

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WORK EXPERIENCE

Databricks

Sr. Software Engineer, Cluster Team

2021-2025

- Engineer on the Cluster Manager team, the compute control plane underlying all of Databricks, which launches tens of millions of VMs across 50+ cloud regions daily and supports \$4B+ in annual revenue.
- Took on the team's largest bottleneck - database instability - driving a multi-quarter effort with the storage team that reduced weekly region-wide outages to effectively zero, directly unblocking the Zero Downtime Cluster Manager project and freeing investment to shift to other strategic priorities.
- Led investigation + resolution of production incidents spanning multiple systems: feature flag inefficiencies causing regional degradation, character encoding issues in Kafka/CDC pipelines causing data corruption, schema migration tooling causing prolonged database lockups
- Made query performance bounded and deterministic by formalizing and enforcing query standards - using secondary indexes and pagination, bounding rows scanned to $O(\text{limit})$ - reducing PMAX query time from minutes to under 1 second; led migrations to MySQL 8.0 and to TiDB
- Designed and prototyped a dataplane worker health monitoring system for detecting and auto-remediating unhealthy nodes, later productionized and shipped by the team
- Owned schema design and data-access review across the Compute Fabric org; gave multiple org-wide talks on system internals and incident analysis; onboarded the Amsterdam engineering team through in-person knowledge transfer and ongoing consultation

Mod9 Technologies

Software Engineer

2018-2021

- Led development of the company's core real-time speech recognition server in C++/Boost
- Built parallelized recognizer capable of transcribing audio up to 400× faster than real time
- Implemented new speaker diarization (segmentation) system, using agglomerative clustering and HMM-GMM models improving diarization error rate from 15% to 4%.
- Trained HMM-DNN models to transcribe English and Spanish using Kaldi.

SKILLS

Languages: Java, C, C++/Boost, Python, Go, Scala, SQL

Infrastructure: Distributed systems, Kafka, MySQL, TiDB, Docker, Kubernetes, Kaldi

EDUCATION

University of California, Berkeley

B.A. Double Major: Applied Math & Computer Science; Regents & Chancellor's Scholarship

London School of Economics

Summer Abroad: International Relations & Cyberlaw